Adapting Segment Anything for Sentinel-1 Water Mapping with Automatic Prompts

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1. INTRODUCTION

Sentinel-1 radar maps surface water through the clouds of extreme rainfall, as in the 2017 coastal El Niño floods in Piura, Peru, which motivate this work. On the hand-labelled chips of Sen1Floods11 [1], the best radar-only result we found is a global IoU of 0.676 with DeepLabv3+ [2]; about 0.72 has been reported with an optical teacher [2] or with auxiliary data and additional weak labels [3]. The Segment Anything Model (SAM) [4] needs prompts and was pretrained on natural images, far from SAR [5]. Low-rank adaptation (LoRA) [6] has helped it on other SAR data [7], and thresholding SAR to prompt SAM has been proposed [8], but in a non-exhaustive search we found no published result for SAM on these chips with radar only.

We ask whether SAM ViT-B can map water with automatic prompts, and what must be adapted. Our contribution is a pre-registered, stepwise adaptation of SAM, evaluated once on held-out data against a strong threshold and a U-Net trained from scratch.

2. METHODOLOGY

Data. Sen1Floods11 v1.1 [1] hand-labelled chips (512 × 512 px, VV/VH in dB) with the official splits. A predeclared rule excludes chips with more than 95% unlabelled pixels, leaving 250 training, 86 validation, 89 test and 15 Bolivia chips. Test shares its regions with training; Bolivia is unseen. One test chip has no radar data and stays as an empty prediction for every method.

Automatic prompts. Otsu thresholding [9] on VH over sensor-valid pixels, followed by a radius-1 opening, gives the baseline and the prompts: three positive and two negative points at the maximum distance to the mask border. Labels are used only for evaluation.

Stepwise adaptation. (1) Tune only the mask decoder with a frozen, cached encoder. (2) Replace per-chip percentile stretching with fixed dB ranges computed on training data. (3) Add LoRA (rank 8, $\alpha = 16$) to the 12 encoder blocks [10] and train it with the mask decoder on [VV, VH, VV−VH] input (4.35 M trainable parameters, 15 epochs, AdamW, learning rate 1e-4, BCE + Dice, random D4 augmentation, seeds 0–2). Hyperparameters were fixed in advance, with no search. The test candidate, fixed before opening test, is the three-seed ensemble with D4 test-time augmentation (TTA).

Baselines and evaluation. Otsu; a U-Net trained from scratch [11] on the same input (7.76 M parameters, 100 epochs); and a fixed VH threshold of −23.3 dB with a 5 × 5 median filter, chosen on training data as in [12] and added after test was opened. Global IoU (TP, FP and FN pooled over chips) is the primary metric and mean per-chip IoU the complementary one, with paired bootstrap 95% intervals over chips and Wilcoxon tests.

3. RESULTS AND DISCUSSION

Automatic prompting fails. Pretrained SAM reaches a global IoU of 0.115 on validation against 0.256 for Otsu, close to marking every pixel as water (0.110). In 41 of 86 chips its mask covers more than 90% of the chip, and its predicted IoU does not track true IoU (Spearman $\rho = -0.05$). Prompts taken from labels (0.234) and the best of three masks (0.190) do not beat Otsu, so better prompts alone do not solve it.

Adaptation. Tuning the decoder gives 0.431 ± 0.038 and the fixed dB input 0.472 ± 0.017. Adding LoRA gives 0.601 ± 0.010, and the final ensemble 0.615 (Table 1). These values guided decisions, so they are optimistic.

Held-out results. SAM + LoRA reaches 0.642 on test (95% CI [0.510, 0.748]) and 0.657 on Bolivia ([0.477, 0.784]). It beats the fixed threshold on test (+0.059 [+0.026, +0.101]), but

not conclusively on Bolivia (+0.044 [−0.005, +0.092]). The U-Net wins on test (+0.024 [−0.007, +0.061], Wilcoxon p = 0.011) and SAM on Bolivia (+0.029 [−0.021, +0.069], p = 0.046). With 15 chips from one region, this is a signal rather than proof that SAM generalizes better. Both models are in the range of DeepLabv3+ (0.676, at 16 m) [2]. SAM is less stable: over five seeds, validation IoU is 0.603 ± 0.017 against 0.635 ± 0.004 for the U-Net, hence the ensemble.

Near-dry chips. Mean per-chip IoU is much lower (0.405 on test) because chips with under 1% water are 31% of test chips but 3% of the pixels, and reach only 0.15: large floods are mapped well, while almost dry images still show proportionally large errors. Exploratory: a JRC permanent-water input to the U-Net barely helped on test (+0.007) and hurt on Bolivia (−0.111 [−0.187, −0.046]).

Method	Valid	Test	Bolivia	Prec.	Recall
Otsu + opening	0.256	0.263 (0.235)	0.467 (0.427)	0.300	0.676
Pretrained SAM, automatic prompts	0.115	—	—	—	—
+ tuned mask decoder†	0.431 ± 0.038	—	—	—	—
+ fixed-range dB input†	0.472 ± 0.017	—	—	—	—
+ LoRA; VV, VH, VV−VH input†	0.601 ± 0.010	—	—	—	—
Fixed VH threshold + median‡	0.539	0.582 (0.314)	0.613 (0.371)	0.812	0.673
U-Net, ensemble + TTA	0.642	0.666 (0.421)	0.628 (0.397)	0.814	0.785
SAM + LoRA, ensemble + TTA	**0.615**	**0.642 (0.405)**	**0.657 (0.438)**	**0.836**	**0.734**

Table 1. Global water IoU (mean per-chip IoU in parentheses) on Sen1Floods11. Test and Bolivia were evaluated once; precision and recall are on test. †Mean ± SD over 3 seeds. ‡Threshold chosen on training data, added after test was opened.

4. CONCLUSIONS

Threshold-derived prompts alone do not make pretrained SAM useful for SAR water mapping. Adapting its input, encoder and decoder reaches 0.642 on test and 0.657 on an unseen region, with radar only and 250 training chips. However, a U-Net trained from scratch beats it within the training regions and is more stable, and a fixed threshold comes close on the unseen region. The evidence is limited by wide intervals, one unseen region and no Peruvian data. Future work: a Piura 2017 test against Copernicus EMS maps, optical information during training, and foundation models pretrained on SAR.

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